

Ganesh Chintala

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PROFESSIONAL SUMMARY

Innovative AI/ML Engineer and Data Scientist with over 5 years of experience designing, deploying, and scaling advanced machine learning and Generative AI solutions across the healthcare, financial, and insurance sectors. Specialized in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and predictive modeling using PyTorch, TensorFlow, and AWS. Proven track record of optimizing data pipelines, fine-tuning state-of-the-art architectures (BERT, FLAN-T5), and driving business intelligence to reduce operational costs and enhance decision-making.

SKILLS

- Languages:** Python, SQL, PySpark, Bash
- Generative AI & LLMs:** Amazon Bedrock, Claude, Llama, BERT, FLAN-T5, RAG, Agentic AI, Model Context Protocol (MCP), Prompt Engineering, PEFT/LoRA, Semantic Search, LiteLLM
- Machine Learning:** Supervised/Unsupervised Learning, Transformers, NLP, Time-series Forecasting, XGBoost, Random Forest
- Frameworks & Libraries:** Scikit-learn, PyTorch, Hugging Face Transformers, LangChain, AutoGen, FastAPI, Pandas, NumPy, Matplotlib, RAGAS
- Tools & MLOps:** MLflow, Docker, Apache Airflow, Jenkins, Git, CI/CD, Langfuse, AWS Step Functions, Amazon ECS Fargate, CloudWatch
- Cloud & Data Platforms:** AWS Bedrock, SageMaker, S3, Glue, EMR, Redshift, Kinesis, OpenSearch Serverless, Step Functions, ECS Fargate, Databricks
- Databases:** PostgreSQL, Amazon Redshift, Hive, Delta Lake, Pinecone, Amazon OpenSearch Serverless, Amazon S3

WORK EXPERIENCE

AI/ML Engineer | UnitedHealth Group, USA | Jan 2025 – Present

- Architected a scalable Generative AI adjudication platform using Model Context Protocol and Amazon Bedrock, increasing straight through processing by 35% and reducing manual review by 45%.
- Engineered an event driven orchestration layer using AWS Step Functions and FastAPI on Amazon ECS Fargate to execute concurrent AI agents and reduce transaction processing latency.
- Designed a Retrieval Augmented Generation pipeline using Amazon OpenSearch Serverless to vector search complex regulatory policies, successfully grounding model reasoning and effectively mitigating critical hallucination risks.
- Integrated LiteLLM as a centralized gateway to dynamically route requests across Bedrock foundation models, implementing automated retries, reliable fallback strategies, and granular cost tracking across applications.
- Established enterprise AI observability by combining deterministic Python validations with language model confidence scoring, utilizing Langfuse to trace prompt latency and tool invocations for complete auditability.

Data Scientist | TATA Consultancy Services (Wells Fargo), India | Dec 2021 – Aug 2023

- Built and productionized a fraud detection ensemble (XGBoost + Isolation Forest) over millions of daily card transactions using PySpark on Databricks, improving fraud recall by 18% while reducing false-positive alert volume by 22% and shrinking the analyst review queue.
- Developed credit risk / probability-of-default models benchmarked against logistic-regression scorecards, validating with AUC, KS statistic, and PSI, and authored model documentation that cleared independent model risk validation under SR 11-7 governance standards.
- Fine-tuned BERT-based classifiers for sentiment and intent detection across customer complaint narratives and call transcripts, automating triage of ~50K records monthly at 0.91 F1 and cutting manual tagging effort by 60%.
- Engineered 200+ reusable features through Apache Airflow DAGs and PySpark transformations on Hive/Delta tables, embedding automated data-quality and schema-drift checks that reduced pipeline failures and rework across downstream models.
- Standardized model deployment using MLflow for experiment tracking and registry with Docker-packaged inference services, reducing model handoff-to-production time from several weeks to a few days.

Data Analytics Engineer | Maxgen Technologies, India | Jan 2020 – Dec 2021

- Developed AWS Glue ingestion pipelines using Python to integrate policy, claims, customer, and premium data from Oracle, SQL Server, Salesforce CRM, and external files into Amazon S3.
- Built scalable PySpark and Spark SQL ETL pipelines on Amazon EMR to transform raw insurance data into curated Silver and Gold datasets, improving data quality and reducing batch processing time by 40%.
- Designed star schemas, data marts, materialized views, and optimized SQL queries in Amazon Redshift to accelerate analytics for Claims, Underwriting, and Finance teams.
- Developed Tableau dashboards and KPI reports for claims settlement, policy renewals, premium collections, and branch performance, enabling self-service analytics and reducing manual reporting effort.
- Automated ETL deployments using Git, Jenkins, Apache Airflow, and CloudWatch while collaborating with business stakeholders to deliver governed analytical datasets supporting enterprise reporting.

EDUCATION

Masters in Data Science | GPA: 4.0/4.0 Lewis University, Romeoville, IL | Jan 2024 – Dec 2025

Bachelors in Electrical and Electronics Engineering | GPA: 3.0/4.0 JNTUK, Kakinada, India | Aug 2017 – May 2021

PROJECTS

- RAG-Based Recommendation System:** Designed and evaluated a RAG-based wine recommendation system using LangChain and Pinecone. Integrated RAGAS to measure end-to-end retrieval and generation quality, achieving answer relevancy of 0.88, faithfulness of 0.60, and context recall of 0.50.
- LLM Optimization for Dialogue Summarization:** Optimized FLAN-T5 for dialogue summarization using PEFT (LoRA) and instruction tuning. Improved ROUGE-1 scores by 13.16% over the base model while achieving near fully fine-tuned performance with 10× lower compute overhead.
- Agentic Data Analyzer System:** Engineered an Autogen-based system where AI agents autonomously write and execute Python code in Docker, enabling conversational Exploratory Data Analysis (EDA), visualizations, and immediate insights from diverse datasets.
- Real-Time Data Streaming Pipeline:** Implemented a real-time taxi fare prediction pipeline using AWS Kinesis Data Streams, Kinesis Data Analytics (Flink), and SageMaker Linear Learner, enabling streaming inference and storing continuous predictions in S3.